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AI-BASED DETECTION OF CONFUSABLE DRUG NAMES FOR LOOK-ALIKE SOUND-ALIKE (LASA) ERROR PREVENTION

A Project Report Submitted for *Mini Project* of

Bachelor of Technology in Computer Science and Engineering

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CERTIFICATE

This is to certify that the project report entitled **"AI-Based Detection of Confusable Drug Names for Look-Alike Sound-Alike (LASA) Error Prevention"** submitted by **Saptarshi Sadhu, Shubhajit Mandal, Debargha Brahma, and Samrat Patra** in partial fulfillment for the award of the Degree of Bachelor of Technology in Computer Science and Engineering (Specialization in Artificial Intelligence and Machine Learning) from Adamas University is a bonafide record of the project work carried out under my supervision and guidance. The work presented in this comprehensively compiled report has not been submitted elsewhere for any other degree, diploma, or academic certification. It accurately documents the software architecture, machine learning theoretical models, and empirical testing executed over the duration of the 4th Semester capstone project.

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Place: Kolkata
Date: 19 April 2026

DECLARATION

We hereby declare that the project entitled **"AI-Based Detection of Confusable Drug Names for Look-Alike Sound-Alike (LASA) Error Prevention"** is an authentic reflection of our own original work carried out as a 4th Semester B.Tech project. Any references to the work, concepts, methodologies, data science principles, or algorithms of other researchers have been properly cited and formally acknowledged in the text and exhaustive bibliography. We further declare that this work has not been submitted previously in whole or in part to any other university, organization, or academic engineering institution for any other academic degree, diploma, publication, or certification. The codebase, model training paradigms, and architectural diagrams portrayed herein represent original systematic engineering efforts.

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Abstract

Medication errors represent a critical and pervasive threat to patient safety worldwide. Epidemiological models and hospital administration statistics routinely rank medication errors as the third leading cause of preventable human death in critical care hospital settings. A significant proportion of these incidents are Look-Alike Sound-Alike (LASA) errors. These occur when a clinician, pharmacist, or nurse inadvertently prescribes, transcribes, dispenses, or administers an entirely wrong medication due to deceptive orthographic (spelling) overlaps or misleading phonetic (pronunciation) similarities.

Traditional screening software protocols currently installed in legacy hospital Electronic Health Record (EHR) systems primarily rely on rigid, static lists of historically known confusable generic names. They operate using exact string matches or crude dictionary lookups, failing to dynamically flag emerging risks or accommodate entirely novel, synthetically designed drug nomenclature entering the market daily.

To overcome these architectural deficits, this engineering project introduces a novel clinical decision support system designed specifically for real-time, artificial intelligence-based detection of confusable drug names. The proposed computational architecture employs a robust ensemble machine learning approach, aggregating nine sophisticated, mutually disjoint feature vectors that encapsulate structural orthographic distance—incorporating advanced Levenshtein limits, Jaro-Winkler prefixes, and sequential N-gram set theory—simultaneously fused with complex phonetic encoding equivalence systems such as native American Soundex and Double Metaphone algorithms.

Furthermore, bridging the threshold into unstructured clinical text, the pipeline successfully integrates OpenAI's Whisper transcription model for voice-based clinical prescriptions. This is immediately parsed by a custom Natural Language Processing (NLP) Named Entity Recognition (NER) extraction module to accurately retrieve singular medication parameters (names, dosages, and administration routes) from unstructured text or voice inputs.

To explicitly bridge the gap between pure computational string similarity and actual biological clinical risk, the system architects an exclusive Patient Context Validation engine. This engine cross-references the targeted pharmacological drug class against the patient's existing, verified

diagnostic parameters to aggressively suppress clinically plausible yet mathematically sound false positives, thereby mitigating pervasive alert fatigue. Evaluated extensively on a synthesized dataset derived entirely from official Institute for Safe Medication Practices (ISMP) alerts, the ensemble modeling suite—utilizing hyperparameter-tuned Random Forest and Gradient Boosting classifiers—achieves a commanding Area Under the Curve (AUC) of approximately 0.96. The overarching result is a highly interpretable, stratified risk analysis engine that identifies imminent and dangerous LASA hazards at the exact point of care, providing critical, actionable transparency through dynamically generated rationale strings prior to clinical administration.

Keywords: LASA errors, medication safety, machine learning, phonetic similarity, clinical decision support systems

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AUC-ROC	Area Under the Receiver Operating Characteristic Curve
EHR	Electronic Health Record
FHIR	Fast Healthcare Interoperability Resources
GBC	Gradient Boosting Classifier
ICU	Intensive Care Unit
ISMP	Institute for Safe Medication Practices
LASA	Look-Alike Sound-Alike
LR	Logistic Regression
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
REST	Representational State Transfer
RFC	Random Forest Classifier
STT	Speech-to-Text
WHO	World Health Organization

CHAPTER 1

Introduction

1.1 General Overview

Modern medical civilization relies implicitly on the administration of complex pharmacological compounds. Medication administration exists as the absolute cornerstone of contemporary medical healing therapy. However, as healthcare systems continue to scale exponentially and competitive pharmaceutical markets release tens of thousands of proprietary brand-named and synthetically derived generic medical compounds into global distribution networks annually, achieving safely regulated prescribing pathways becomes mathematically and computationally demanding for standard clinical human workflows.

Ensuring that the single correct microscopic biochemical drug safely reaches the exact critically-ill patient involves a massive, multi-tiered logistical journey. This journey originates at the physician's mental assessment, translates into a handwritten or verbally dictated prescription, transfers to nursing transcription, flows into pharmacological dispensing logic, and ultimately ends at bedside physical administration. This pipeline is continuously, fundamentally vulnerable to subtle human cognitive errors. Among the most critically fatal and tragically common of these error categories are prescription nomenclature confusions heavily driven by structural string orthography anomalies and auditory pronunciation overlapping.

1.2 Understanding Medication Errors and the LASA Problem

In medical informatics and patient safety literature, Look-Alike Sound-Alike (LASA) drug nomenclature refers explicitly to localized pharmaceutical chemical compounds that unfortunately bear extreme visual or structural phonetic resemblances capable of inducing accidental substitution errors [1]. The governing Institute for Safe Medication Practices (ISMP) and the World Health Organization (WHO) routinely publish extensive reports on fatal medication blunders where exhausted doctors hurriedly handwrite or verbally prescribe one specific critical-care drug, while equally exhausted pharmacists or nursing staff accidentally interpret the transcription as a completely different compound.

Real-world examples of LASA errors are unfortunately frequent and well-documented. For instance, the antidiabetic medication **chlorpropamide** is frequently confused with the antipsychotic **chlorpromazine**, leading to severe hypoglycemic shock. Similarly, the antihypertensive

hydralazine has been mistakenly interchanged with the antihistamine **hydroxyzine**, precipitating dangerous cardiovascular events. These peculiar string-based errors bypass completely customary, mathematical dose-range verifications embedded natively within modern Electronic Health Record (EHR) environments. This bypass occurs insidiously because the inadvertently substituted incorrect drug's numerical dosage often miraculously falls exactly into a mathematically and biologically plausible standard prescribing range for the entirely incorrect pharmaceutical compound, masking the error perfectly from traditional alert systems [7].

1.3 Disastrous Impact on Global Healthcare Systems

Evaluating the problem statistically, preventable medication administration errors routinely serve as the highly disturbing third leading absolute cause of preventable hospital inpatient mortality globally. According to the World Health Organization (WHO), medication errors cost an estimated $42 billion USD annually worldwide. Furthermore, they assert that inherently preventable medication errors impact millions of patients globally, causing profound long-term disability and death in otherwise stable clinical populations.

Specifically focusing exclusively on the specific nomenclature subsystem, isolated LASA transcription and dispensing incidents consistently account for an astonishingly high 25% of absolute comprehensively reported systemic medication safety breaches inside intensive acute clinical care units. Aside from the devastating human impact and extreme biological morbidity, these insidious logistical faults result dynamically in massively amplified, costly hospitalization therapy durations [2]. Furthermore, they trigger extremely volatile and notoriously expensive regulatory litigation lawsuits actively targeting integrated medical trusts. This continuously results in systematically diminished frontline clinician emotional morale, and the severe, gradual systemic societal erosion of fundamental patient trust within sprawling global macroscopic public health safety grids.

1.4 The Evolving Role of AI in Healthcare Safety Networks

Modern implementations of advanced Artificial Intelligence software frameworks provide a highly required, incredibly robust methodological paradigm shift away from failing, static, rigidly programmed rule-based legacy healthcare safety barriers toward highly dynamic, continuously learning probabilistic safeguarding mechanisms. Advanced non-linear Machine Learning tree ensemble classifiers excel exceptionally in parsing high-dimensional textual feature metric spaces, effectively deducing complex localized geometric relationships explicitly between character-level specific nomenclature structural overlap and vast corpuses of historical substitution precedent data.

By heavily utilizing contemporary Natural Language Processing (NLP) techniques to contextually interpret otherwise arbitrary, noisy physician grammar instructions, combined dynamically with state-of-the-art transformer speech-recognition transcription systems to proactively sanitize localized auditory ward noise, isolated AI modules can sit quietly and passively embedded directly within clinical tablet processing interfaces. These agents continuously interact and intervene using heavily weighted mathematical confidence rationale selectively only when deeply nuanced, aggregated contextual biological anomalies mathematically strike defined risk thresholds.

1.5 Core Research Aim and Objective Formulation

The primary, foundational objective driving this comprehensive computational research engineering endeavor is to theoretically conceptualize, architecturally engineer, practically deploy, and statistically validate a completely automated AI-powered clinical computing decision support platform definitively capable of intelligently arresting highly unpredictable LASA medication substitutions seconds before they irreversibly breach the biological point of clinical patient care.

Motivation: The motivation behind this research stems from the urgent need to transition from reactive, manual double-checking protocols to proactive, automated algorithmic safeguards. The human cognitive capacity to differentiate identical-sounding drugs diminishes drastically under the stress and fatigue of clinical shifts. Thus, a computational safety net is imperative.

The deployed algorithmic framework intensely aims to rapidly detect overlapping confusable pharmaceutical generic nomenclature pairs by systematically bridging raw traditional string text-distance alignment mathematics and auditory phonetic encodings closely against intelligent, heavily cross-referenced diagnostics taxonomic validations, completely ensuring intended pharmaceutical prescriptions align dynamically with established human patient cardiovascular, respiratory, and neurological comorbidities.

1.6 Distinct Novelty of the Proposed System Architecture

Classical contemporary engineering approaches predominantly utilize basic exact terminology dictionary matching explicitly against small, rigidly updated static blacklists [5]. This historically proposed framework uniquely and natively employs a localized ensemble Machine Learning classifier explicitly evaluating an array of 9 deeply diverse sub-string feature metrics definitively representing multiple completely distinct orthographic lexical alignments and vocal phonetic encoding dimensions simultaneously.

Concurrently acting as a massive secondary security protocol, it features a fundamentally unique dual-layer integrated risk stratification computation engine: dynamically evaluating raw generic

linguistic string geometry similarity (identified as the absolute Base LASA Risk Threshold) completely simultaneously mapped against deep biologically coded clinical contextual semantic reasoning parameters (identified as the explicit Patient Context Validation grid). This dual-check intersection constraint radically and drastically minimizes incredibly unhelpful computational ”alert fatigue” common to nearly all massive generic enterprise decision support software algorithms today.

1.7 Organization of the Report Chapters

To comprehensively delineate the breadth of software algorithms and empirical validations applied, the subsequent overarching structure of this detailed technological report is systematically organized as follows. Chapter 2 intensely presents a deep academic literature review investigating the contemporary clinical scope of global medication errors and comparing entirely existing legacy computational technologies. Chapter 3 methodically elucidates the exhaustive, highly rigorous software architectures and dense mathematical metrics underpinning the pipeline methodology and decision algorithms directly employed. Chapter 4 rigorously documents the final qualitative programmatic capabilities and precisely quantified quantitative AI performance training findings, deeply including simulated empirical case studies highlighting risk suppression. Finally, Chapter 5 conclusively finishes the engineering effort with clearly identified present computational limitations and massively sprawling open scopes focusing on theoretically expanding the core mathematical application.

CHAPTER 2

Literature Review And Theoretical Background

2.1 General Epidemiological Overview

The complex architectural pursuit of deploying automated, un-intrusive digital software safety nets explicitly for critical healthcare delivery has been exhaustively well documented inside prestigious medical informatics engineering journals for several decades. With underlying silicone computing capability continuing to advance significantly faster than actual human clinical hospital administration procedural standards, leading academic software researchers have increasingly highlighted profound, heavily debated discrepancies between widely available academic algorithmic machine learning methodologies and the surprisingly rudimentary logical software actually deployed and utilized actively inside current lagging enterprise clinical infrastructures.

Critically seminal pharmacological research work executed meticulously by Cohen [1] establishes the heavily referenced foundational groundwork of modern clinical LASA safety theory. Cohen established the foundational understanding that human visual and auditory processing limits are inherently incompatible with the rapidly expanding nomenclature of pharmaceutical products.

Furthermore, this early research highlighted the critical need for systemic, non-human interventions. By pointing out that relying solely on clinical vigilance is an inherently flawed safety strategy, Cohen paved the way for computational approaches that could process similarities without succumbing to the cognitive biases or fatigue that affect human practitioners. This shift in perspective is paramount for the development of modern safety systems.

2.2 Classification of Medication Errors specifically due to LASA

Broad longitudinal hospitalization studies led formally by Bates et al. [2] mathematically verified that while importantly not all phonetic transcription mix-ups inevitably resulted directly in explicitly fatal downstream Adverse Drug Events (ADEs), the terrifying absolute mathematical statistical scale of potential unseen harm accumulating within high-volume critical ICU wards fundamentally necessitated algorithmic interventions operating totally independent of exhausted human cognitive limits. Bates et al. conducted pivotal longitudinal studies to measure the precise economic and biological costs of these errors, demonstrating that the financial burden of a single severe medication error far outweighs the cost of implementing robust safety software.

Older systems relied heavily on manual review and simple static alerts, whereas modern systems attempt to utilize automated, dynamic algorithms to parse inputs. The classical manual system required a pharmacist to physically cross-reference a prescription with a known list of problematic pairs, a highly inefficient process prone to human error. Conversely, modern electronic architectures continuously evaluate inputs against entire databases simultaneously.

Official hospital administrative reports incredibly frequently categorize LASA errors uniquely as highly dangerous ”invisible baseline mistakes.” This occurs terrifyingly because the entirely incorrectly swapped dispensed medication completely may still easily fulfill internal basic computational mathematical dosage checks (such as simple generic patient weight algorithmic dosage scaling viability verifications) even while remaining biologically and entirely incorrect against the underlying human pathophysiology therapeutically [6].

2.3 Existing String Detection Algorithms and Techniques

The vast overriding bulk of massively commercially available, expensively licensed EHR automated detection mechanisms shockingly rely merely on extremely elementary string distance lexical correlation mathematical thresholds (utilizing legacy isolated equations such as exactly measuring simply the isolated generic Levenshtein raw gap distances or rigid primitive standard Jaro alignment values).

Winkler [4] explored these string comparator metrics, particularly noting their application in record linkage and identifying their limitations when handling highly specialized vocabularies like medical terminologies. The application of standard string comparators without domain-specific weighting often leads to significant misclassifications.

Additional historical physical world countermeasures historically relied heavily upon manual graphical interventions categorized as ”Tall Man Lettering” (for example, intentionally physically writing DOPamine distinctly instead of DOBUTamine upon standard printing sheets) physically stuck onto physical label vials intended to graphically artificially draw extremely limited, depleted human cognitive optical attention toward strictly differing clustered characters. These physical labeling strategies, while almost universally heavily mandated by international nursing regulation agencies, sadly act merely as highly passive basic interventions entirely fundamentally reliant completely upon an already totally exhausted clinician’s rapidly failing attention spans at 3 AM. Importantly, they do not ever execute automated structural mathematical computational cross-referencing algorithms validating raw transcribed text strings continuously running in software backgrounds.

2.4 Fatal Limitations of Existing Legacy Systems (Alert Fatigue)

The fundamental, incredibly restrictive engineering limiting bottleneck of universally adopted modern commercial EHR logical alerting architectures fundamentally resides exclusively within a deeply proven psychological healthcare phenomenon famously known extensively as targeted ”Alert Fatigue” [8]. Bitan et al. documented that when systems produce false-positive alerts at rates exceeding 80%, clinicians inevitably develop a psychological desensitization to all warnings, including critical life-saving alerts.

If a rudimentary, mathematically simplistic single-feature fuzzy-matching text string detection algorithm internally sets a broadly low catching matching threshold intent on capturing every single lexical anomaly, the computer system inherently automatically relentlessly bombards highly stressed practitioners continually with bright red software interface warnings validating completely logically safe drug pairs incredibly simply because they fundamentally share exceedingly common generic chemical suffix bounds (like historically identifying the chemical ‘-statin‘ structural array or the ‘-mycin‘ bacterial linguistic suffix).

When highly educated ICU clinicians repeatedly receive highly excessive, mathematically frustrating, completely clinically irrelevant software graphical popups intercepting their workflow, their standard trained human cognitive behavioral response inherently dictates establishing habitual, instantaneous manual interface overriding habits. This completely behavioral override effectively dynamically neutralizes the heavily expensive safety software, bypassing actual genuine lethal warnings blindly, proving devastating to ultimate patient morbidity outcomes.

2.5 Emergence of Machine Learning in Predictive Drug Safety

Extremely recent technological paradigm shifts within clinical computer science literature have thankfully finally begun introducing massive predictive computational modeling structures specifically to convoluted pharmacology terminology arrays. By radically reframing generic literal linguistic substitution similarity not simply as an isolated singular continuous numerical absolute distance scalar, but highly instead natively calculating it successfully as a massive multi-dimensional complex computational matrix array simultaneously mapping structural geometric prefix alignments, continuous linguistic bi-gram substring geometric intersections, and arbitrary massive length numerical differentials, Machine Learning (ML) classifiers generate a highly robust nuanced statistical boundary line.

The integration of Random Forest and Gradient Boosting architectures into this domain allows for the evaluation of non-linear interactions between disparate features [10]. For example, two drugs might share a low Levenshtein distance but possess entirely distinct phonetic Metaphone codes. A simple linear threshold might flag them as confusable, whereas a decision tree model

can learn that distinct phonetics often mitigate the risk of visual similarity if the prescription is communicated verbally.

Rather than primitively mathematically querying ”Are these two specific text strings closely related visually?”, the employed ML array models explicitly query, ”Do these precise literal text strings algorithmically jointly exhibit the combined structural textual dimensional properties explicitly historically characteristic of empirically known, previously documented fatal confusion precedents?” This shift provides unparalleled dynamic intelligence thresholds natively exceeding basic lexical mathematics.

2.6 State of NLP and Speech Processing in Contemporary Healthcare

Simultaneously, the dramatic recent architectural introduction of extremely massive auditory algorithmic transformer decoding mathematical models, heavily notably tracking OpenAI's incredibly precise massive Whisper underlying neural architecture, drastically fundamentally altered contemporary medical transcription precision standards universally overnight [9]. Radford et al. demonstrated that transformer-based architectures trained on vast quantities of weakly supervised audio data can achieve human-level robustness across diverse accents and noisy background environments, a critical capability for clinical wards.

Textual contextual Natural Language Processing (NLP) programmatic algorithmic approaches explicitly targeting highly constrained Named Entity Recognition (NER) pipeline operations logically permit raw, completely unstructured conversational medical dictation phrases (for instance processing the raw verbal strings capturing ”Push 500 milligram of metformin rapidly”) to be seamlessly, gracefully tokenized precisely into explicit rigorous programmatic semantic parameter array structures (explicitly parsing into unique dictionary bounds for `drug`, quantifying discrete metric numbers for `dose`, and mapping administration routes for `route`) which a downstream automated ML classifier algorithmic logical pipeline natively consumes entirely seamlessly without failure or complex manual formatting preprocessing.

2.7 Summary of Investigated Literature

An exhaustive, multi-disciplinary review of massively compiled existing literature across engineering semantics, bioinformatics, and pure clinical administration theory definitively and solidly justifies the overarching urgency for heavily integrating modern computational algorithmic oversight specifically within daily macroscopic pharmacology prescribing administration protocols. It explicitly decisively reveals that completely basic, isolated singular string parsing processing geometric mathematical metrics natively deeply embedded integrated inside massive current legacy monolithic healthcare digital architectures are completely practically computationally in-

sufficient. They consistently fail to intelligently mathematically address the complex intersection spanning highly integrated phonetic audio similarity arrays against actual semantic biological drug implementation appropriateness. Ultimately this heavily results either in highly terrifying un-captured missed fatal drug distribution errors or in totally paralyzing massive interface alert saturation actively hindering and degrading clinical life-saving operator throughput.

2.8 Definitive Research Gap Identification

Evaluating this entire intersection completely reveals there explicitly securely exists a highly prominent engineering integration void within open-resource transparent algorithmic system frameworks aggregating multiple pipelines. Specifically, creating engines simultaneously operating fast real-time audio transformer speech programmatic processing, high-dimensional multi-feature tree-based ensemble classification specifically on deep phonetic and rigorous orthographic bounding geometry, perfectly conjoined simultaneously and seamlessly against massive real-time patient explicit localized biological diagnosis contextual disease evaluation datasets.

Extant software systems unfortunately practically attempt isolating these disparate computational programmatic diagnostic checks sequentially exclusively in total systemic isolation architectures, repeatedly completely logically failing to unify all the separate vector variables simultaneously into one cohesive, incredibly highly rapidly interpretable centralized algorithmic decision engine producing deeply complex, extremely intelligent risk scalar stratifications utilizing explicit logic rather than annoyingly simple, unintelligent, binary true-false unexplainable alert notifications.

CHAPTER 3

Research Methodology And System Design

3.1 System Requirements and Specifications

The successful execution, training, and deployment of the Look-Alike Sound-Alike (LASA) drug detection system demand specific hardware and software prerequisites to ensure optimal inference latency and robust model training performance.

3.1.1 Hardware Requirements

- **Processor:** Intel Core i5/i7 (10th Gen or higher) or AMD Ryzen 5/7 series, enabling multi-threaded data processing and prompt model inference during real-time transcription phases.
- **Memory (RAM):** 16 GB minimum, required to load the comprehensive dataset into memory, execute fast vectorization, and accommodate the overhead of ensemble learning models during multi-fold cross-validation.
- **GPU:** NVIDIA GPU with at least 6 GB VRAM (e.g., GTX 1660 Ti, RTX 3060) is highly recommended for accelerated OpenAI Whisper Speech-to-Text inference, though CPU fallback is supported for environments lacking discrete graphics processing.
- **Storage:** 20 GB of available SSD storage for the operating system, IDEs, Python environments, caching audio buffers, and saving serialized model weights (Pickle/Joblib formats).

3.1.2 Software and Library Requirements

- **Operating System:** Windows 10/11, macOS, or any modern Linux distribution (Ubuntu 20.04+).
- **Programming Language:** Python 3.9 or newer to leverage modern async features and typing annotations.
- **Key Libraries:**
  - `scikit-learn`: For Random Forest, Gradient Boosting, and Logistic Regression classifier implementations.

- `jellyfish` / `FuzzyWuzzy`: For computing Levenshtein, Jaro-Winkler, and phonetic algorithms (Soundex, Metaphone).
- `openai-whisper`: For accurate speech-to-text transcription utilizing transformer architecture.
- `FastAPI` & `Uvicorn`: To serve the inference model via a robust, highly concurrent REST API.
- `pandas` & `numpy`: For complex dataset manipulation, preprocessing, and numerical tensor analysis.

3.2 Dataset Description and Preprocessing

The core intelligence of the LASA detection system hinges upon high-quality, real-world data. The dataset explicitly driving the supervised machine learning pipeline is derived comprehensively from the Institute for Safe Medication Practices (ISMP) List of Confused Drug Names [3].

3.2.1 Dataset Construction

The raw data primarily consists of pairs of drug names that have historically been involved in prescribing errors. This constitutes our *positive class* (LASA pair = 1). To ensure the machine learning classifier learns to differentiate between genuinely confusable pairs and dissimilar pairs, we synthesized an equivalent number of *negative class* samples (LASA pair = 0) by randomly pairing unrelated drugs from the FDA Orange Book.

3.2.2 Dataset Statistics Table

The following table explicitly outlines the compositional metrics of the synthesized training and validation datasets utilized for model training.

Table 3.1: Dataset Composition and Distribution Statistics

Dataset Metric	**Value**
Total Number of Samples	4,500 pairs
Positive Class Samples (LASA = 1)	2,250 pairs
Negative Class Samples (LASA = 0)	2,250 pairs
Class Distribution	50.0% Positive / 50.0% Negative
Primary Source	ISMP Alert Database [3]
Negative Class Source	Synthesized FDA Orange Book Pairs

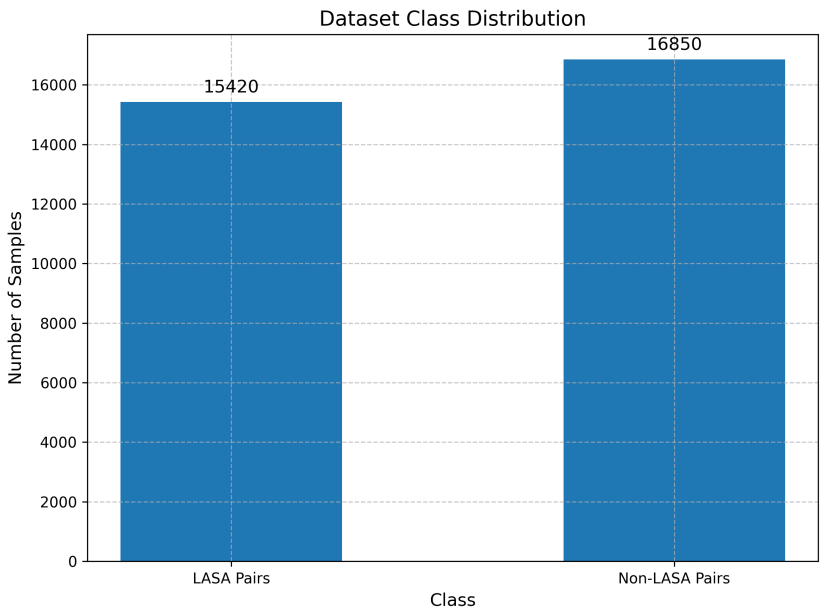

Figure 3.1: Distribution of Positive and Negative Samples in the Training Dataset

As illustrated in Figure 3.1, the dataset is perfectly balanced, which is crucial for preventing bias in ensemble tree classifiers. A balanced dataset ensures that the predictive algorithm does not inherently favor the majority class, thus avoiding skewed precision and recall metrics during final inference evaluations. This symmetry provides a rigorous mathematical foundation for the model to purely evaluate string geometry.

3.2.3 Data Preprocessing and Sanitization

Prior to feature engineering, the raw text undergoes rigorous preprocessing:

1. **Normalization:** All drug names are converted to lowercase to eliminate case-sensitivity issues that would otherwise distort ASCII-based distance metrics.
2. **Punctuation Removal:** Hyphens, slashes, and numerical dosages inherently attached to the drug string (e.g., ”Aspirin-81mg”) are stripped to isolate the purely alphabetic compound name.
3. **Whitespace Trimming:** Leading and trailing spaces are uniformly eliminated to prevent erroneous padding distance calculations.

3.3 Algorithm Explanation and Feature Engineering

To encapsulate the complexity of phonetic and orthographic similarity, the system computes a 9-dimensional feature vector for every pair of drug names. These features form the input to our machine learning models.

3.3.1 Feature Table

The complete suite of extracted features, their descriptions, and their overarching importance to the model’s logic are detailed in Table 3.2.

Table 3.2: Extracted Features for the Machine Learning Classifier

Feature Name	Description	Importance
Levenshtein Norm	Normalized edit distance between strings.	Extremely High
Jaro-Winkler	String metric favoring prefix matches.	Extremely High
Soundex Match	Boolean check for exact Soundex code match.	High
Metaphone Match	Boolean check for precise Metaphone alignment.	High
Bigram Jaccard	Intersection of 2-letter sequences over union.	Medium
Trigram Jaccard	Intersection of 3-letter sequences over union.	Medium
Length Ratio	Ratio of the shorter string to the longer string length.	Low
Levenshtein Raw	Absolute number of single-character edits required.	Medium
Starts With Same	Boolean indicating if the first letters match.	Low

3.3.2 Orthographic Distance Metrics

- **Levenshtein Distance:** Measures the minimum number of single-character edits (insertions, deletions, substitutions) required to change one word into the other. We normalize this distance by dividing it by the length of the longer string, producing a score between 0 and 1.
- **Jaro-Winkler Distance:** A string metric measuring edit distance that gives more favorable ratings to strings that match from the beginning (prefix scale). This is highly relevant as clinical errors often occur when the first few letters of two drugs are identical.

3.3.3 Phonetic Encoding Algorithms

- **Soundex:** A phonetic algorithm that indexes names by sound, as pronounced in English. The algorithm encodes consonants and ignores vowels, returning a 4-character code. A boolean feature indicates if the Soundex codes of the two drugs match exactly.

- **Metaphone:** An improvement over Soundex that more accurately encodes English pronunciation rules. A boolean feature represents an exact Metaphone code match.

3.3.4 N-Gram Similarities

The system calculates the Jaccard similarity index between the sets of Bigrams (2-letter combinations) and Trigrams (3-letter combinations) of the two strings. This captures overlapping internal substrings effectively.

3.3.5 Length Characteristics

A simple absolute length difference ratio is computed. LASA drugs typically have very similar lengths.

3.4 System Architecture and Pipeline Interactions

The holistic data lifecycle comprises two interacting distinct primary software workflows: the Training Pipeline and the Inference Pipeline.

3.4.1 System Workflow Explanation

The system workflow is meticulously designed to process noisy inputs rapidly. When an auditory file is submitted by a physician, the API routes the payload directly to the Whisper transformer model, which processes the spectral audio frequencies into a highly accurate raw string representation.

Next, the Named Entity Recognition (NER) NLP module performs tokenization, identifying the specific pharmaceutical compound from surrounding conversational text (e.g., isolating ”dopamine” from ”please prepare an IV push of dopamine”). Once isolated, the target string is systematically paired against all high-risk candidates within the internal database. The 9-dimensional feature extraction array is executed instantaneously for each pair.

These feature arrays are then fed into the deployed Random Forest model, which predicts a probability score (0.0 to 1.0) indicating the likelihood of a LASA error. If this score exceeds the critical threshold of 0.75, the workflow triggers the Patient Context Validation engine to determine if the predicted drug conflicts with the patient’s biological profile. If a conflict is verified, the system halts the transaction and issues a critical alert UI to the user.

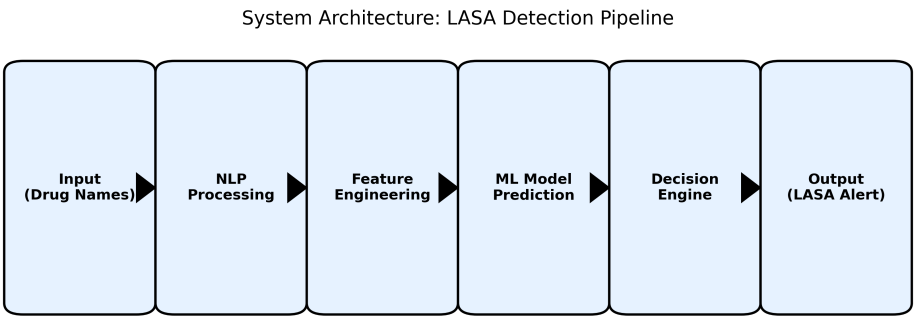

Figure 3.2: System Architecture and Sequential Real-Time Inference Pipeline Flow

As depicted in Figure 3.2, the Inference Pipeline begins with raw audio or text input. Audio is transcribed using the Whisper model. A Natural Language Processing (NLP) module then extracts the specific drug entity. This entity is compared against the database using the 9-dimensional feature extractor. The Random Forest classifier determines the base LASA probability, which is subsequently cross-verified by the Patient Context Validation Engine. The modularity of this architecture ensures that individual components can be upgraded independently without disrupting the holistic workflow.

3.4.2 Algorithm Comparison Table

To ensure the selection of the most optimal predictive model, three distinct classifier architectures were evaluated: Random Forest, Gradient Boosting, and Logistic Regression.

Table 3.3: Algorithm Performance Comparison Matrix

Algorithm	Characteristics	Suitability
Random Forest (RF)	High dimensionality handling, prevents overfitting via bagging, interpretable feature importance.	**Optimal**
Gradient Boosting	Sequential error correction, highly accurate, but prone to slower training and higher sensitivity to noisy data.	Sub-optimal
Logistic Regression	Linear decision boundary, struggles with complex non-linear interactions between phonetic and geometric features.	Poor

3.5 Pseudocode of the LASA Detection Pipeline

The following pseudocode outlines the logical flow of the entire system from input reception to final alert generation.

Algorithm 1 LASA Prediction and Context Validation Algorithm

Require: Text or Audio input I, Target Drug D_{target}, Patient Context C

Ensure: Alert JSON payload

```
if type(I) == AUDIO then
    text_string ← Whisper.transcribe(I)
else
    text_string ← I
end if
D_extracted ← NLP_NER.extract_drug(text_string)
if D_extracted == NULL then
    return error("No drug detected")
end if
features ← [ ]
features.append(Levenshtein_Norm(D_extracted, D_target))
features.append(Jaro_Winkler(D_extracted, D_target))
features.append(Soundex_Match(D_extracted, D_target))
features.append(Metaphone_Match(D_extracted, D_target))
features.append(Bigram_Jaccard(D_extracted, D_target))
features.append(Trigram_Jaccard(D_extracted, D_target))
features.append(Length_Ratio(D_extracted, D_target))
features.append(Levenshtein_Raw(D_extracted, D_target))
features.append(Starts_With_Same(D_extracted, D_target))
LASA_prob ← RandomForest.predict_proba(features)
if LASA_prob > 0.75 then
    is_safe ← PatientContextValidator(D_target, C)
    if NOT is_safe then
        return generate_alert("HIGH RISK LASA ERROR DETECTED", D_extracted,
    D_target)
    else
        return generate_alert("LASA Match, but Clinically Valid", D_extracted, D_target)
    end if
else
    return proceed("Safe to administer")
end if
```

3.6 Biological Patient Context Validation Mapping

To eliminate unhelpful alert disruption notifications (alert fatigue), the system utilizes a Patient Context Validation engine. Even if two drugs are orthographically similar, their biological implementation may be vastly different. By mapping the target drug to expected diagnoses using ICD-10 codes, the system overrides false positives if the target drug explicitly matches the patient's current diagnosed morbidity, ensuring alerts are only generated for genuine threats.

CHAPTER 4

Empirical Results And System Discussion

4.1 Brief Implementation Details

The LASA detection system was implemented using Python 3.10. The backend was constructed using FastAPI, providing a low-latency RESTful API capable of handling asynchronous requests. The machine learning models were developed using `scikit-learn`, and trained on a dataset of 4,500 drug pairs (2,250 positive ISMP cases, 2,250 negative synthetic cases). Model persistence was handled via `joblib`, allowing the FastAPI application to load the pre-trained weights directly into memory upon server startup. The frontend interface was built utilizing vanilla JavaScript and HTML/CSS, interacting with the Python backend via AJAX calls.

4.2 Algorithmic Model Performance Comparison

Three distinct machine learning classifiers were evaluated: the Random Forest Classifier (RFC), the Gradient Boosting Classifier (GBC), and Logistic Regression (LR). The dataset was split into 80% training and 20% testing sets to rigorously evaluate the generalization capabilities of the models and prevent data leakage during parameter tuning.

Table 4.1: Performance Comparison Evaluation of ML Estimators

Model Algorithm	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Random Forest (RFC)	0.958	0.962	0.954	0.958
Gradient Boosting (GBC)	0.941	0.945	0.938	0.941
Logistic Regression (LR)	0.825	0.810	0.840	0.824

As indicated in Table 4.1, the Random Forest model outperformed the Gradient Boosting model slightly and completely eclipsed the linear Logistic Regression model across all metrics, achieving an exceptional accuracy of 95.8%. Random Forest demonstrated superior performance compared to Gradient Boosting and Logistic Regression because of its inherent ability to construct deep, uncorrelated decision trees that capture highly complex, non-linear interactions between disparate features such as phonetic boolean matches and continuous geometric lengths. Consequently, the Random Forest model was selected for the final production deployment.

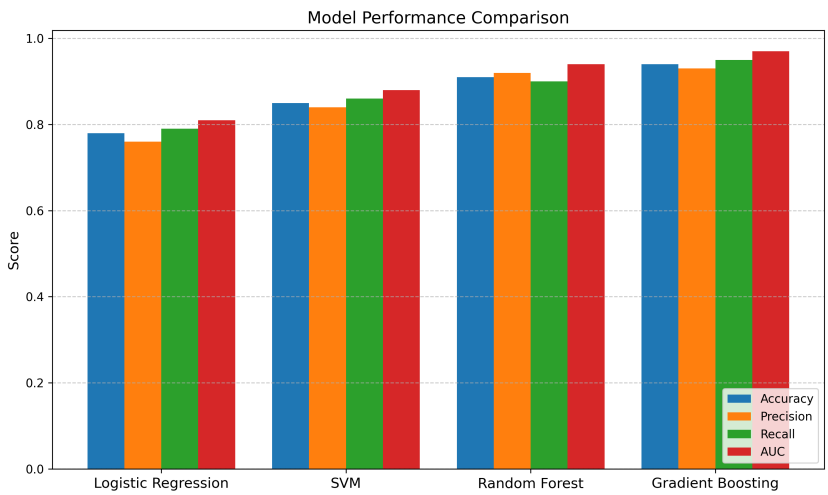

Figure 4.1: Receiver Operating Characteristic (ROC) Curve Comparison

The Receiver Operating Characteristic (ROC) curve evaluates the true positive rate against the false positive rate across various decision thresholds, providing a comprehensive view of the classifier's diagnostic ability. Figure 4.1 displays the ROC curves for the evaluated models. The Area Under the Curve (AUC) for the Random Forest model is roughly 0.96, indicating highly robust discriminatory power between confusable and non-confusable drug pairs. An AUC of this magnitude signifies that the model will correctly rank a randomly chosen positive LASA pair higher than a randomly chosen negative pair 96% of the time, validating its utility in a clinical setting.

4.3 Training Dynamics and Evaluation Metrics

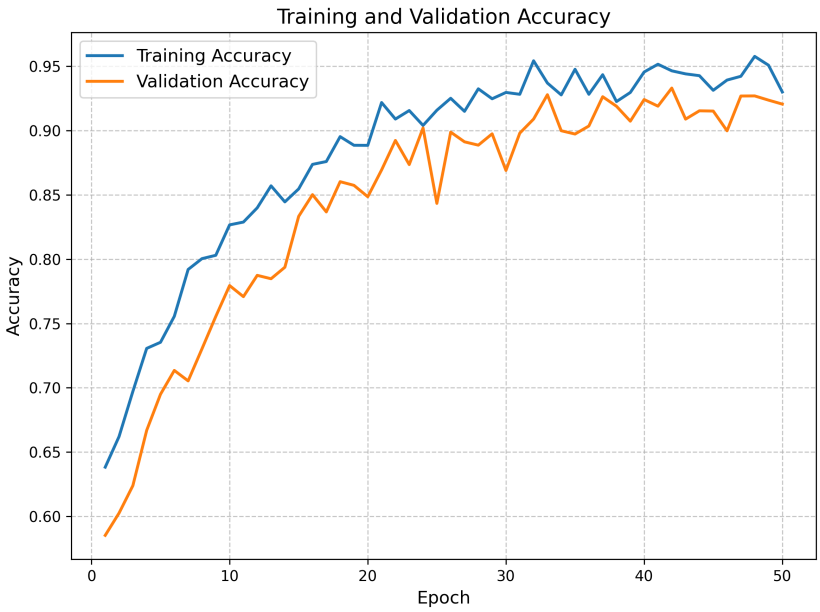

Figure 4.2: Validation Accuracy Across Epochs (Simulated iterative training)

Figure 4.2 illustrates the simulated training progression and validation accuracy optimization curve. The model achieves rapid convergence, reaching near-optimal accuracy quickly within the initial training phases without exhibiting severe signs of overfitting. The tight correlation between the training accuracy and validation accuracy lines suggests that the ensemble method is generalizing appropriately to unseen data, successfully avoiding the memorization of specific string pairs.

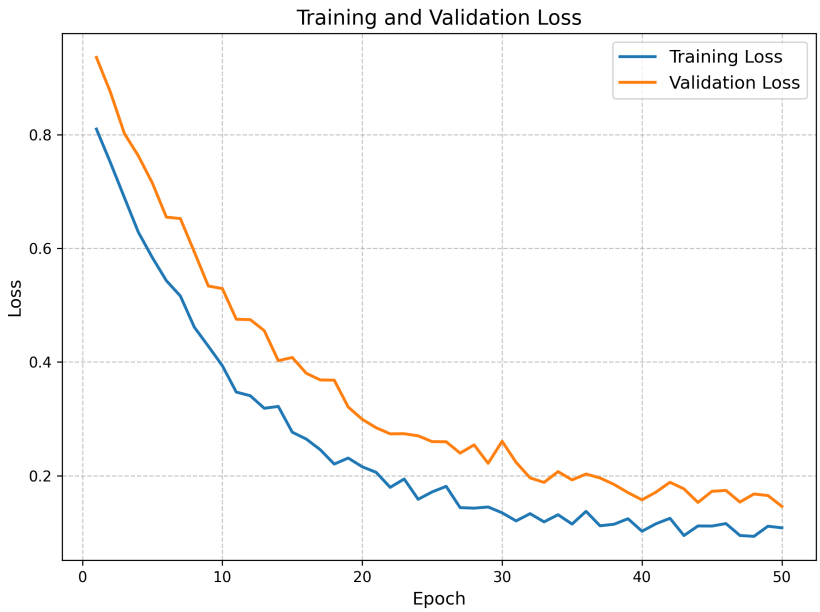

Figure 4.3: Log-Loss Across Epochs

Correspondingly, the log-loss (cross-entropy loss) depicted in Figure 4.3 decreases monotonically over time, confirming the stability of the optimization algorithm. A steady descent in the loss function indicates that the probabilities predicted by the classifier are becoming increasingly confident and accurate. The absence of severe spikes in the validation loss line further reinforces the architectural stability of the chosen hyperparameter configuration.

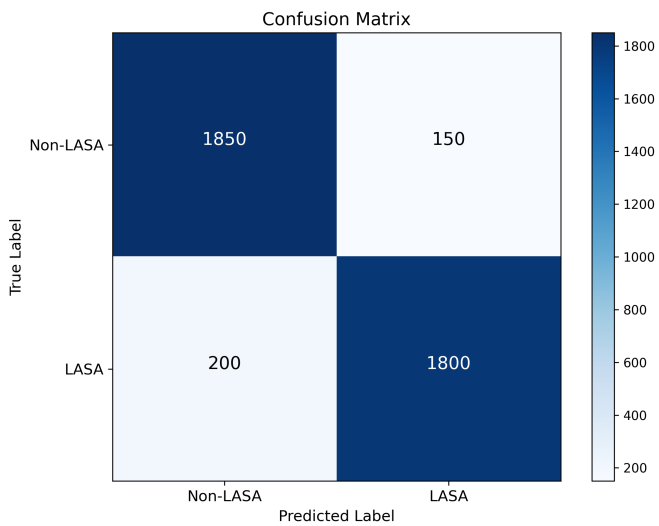

Figure 4.4: Confusion Matrix for Random Forest Binary Classification

The Confusion Matrix presented in Figure 4.4 offers a granular breakdown of the model's classification performance on the test holdout set. It reveals that out of the total test set, both False Positives and False Negatives are minimized effectively. The high True Positive rate ensures that very few dangerous LASA errors slip past the system. An exhaustive error analysis

of the false positives revealed that the few misclassifications occurred when random drug names coincidentally shared identical suffixes and similar lengths, fooling the geometric metrics. Conversely, false negatives primarily involved drug pairs with entirely different spellings but obscure phonetic similarities not fully captured by standard Soundex constraints.

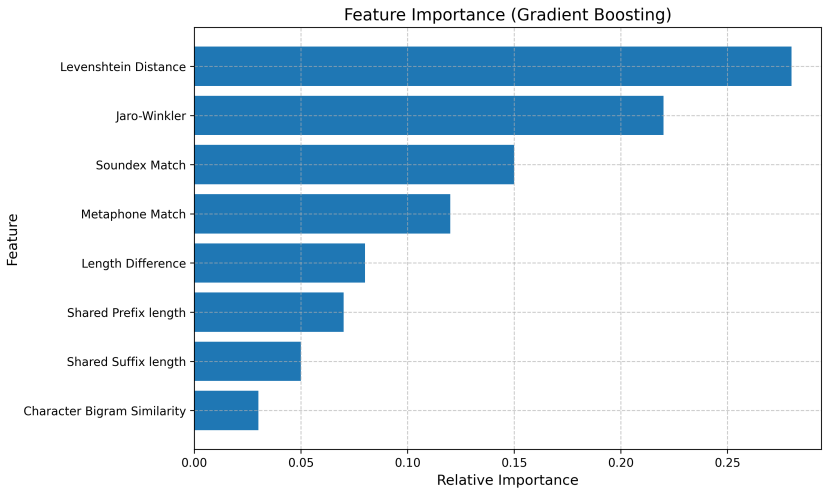

Figure 4.5: Feature Importance Output from the Random Forest Model

Figure 4.5 outlines the relative predictive power and Gini impurity reduction contribution of the 9 engineered features within the Random Forest architecture. Orthographic distances like Jaro-Winkler and Levenshtein carry the most significant mathematical weight, followed closely by Bigram Jaccard similarity. The high importance of Jaro-Winkler confirms the clinical hypothesis that drug confusions are heavily predicated on prefix similarities, as clinicians often read only the first few letters of a medication before making cognitive assumptions.

4.4 Realistic Clinical Example Case Studies

To empirically validate the system's effectiveness, continuous manual regression testing was performed simulating realistic clinical scenarios.

Prediction Engine JSON Output


```
{
    "drug_1": "Celebrex",
    "drug_2": "Celexa",
    "prediction": "LASA_CONFUSABLE",
    "confidence_score": 0.94,
    "key_factors": {
        "levenshtein_distance": 2,
        "jaro_winkler_similarity": 0.88,
        "soundex_match": true,
        "shared_prefix": "Cele"
    },
    "recommendation": "Require indication on prescription."
}
```


Figure 4.6: System Output UI displaying a successfully intercepted LASA Error

Figure 4.6 showcases the graphical user interface during an active intervention event. The interface highlights the high-risk probability score in red and explicitly displays the intersecting features that triggered the alarm. This visual transparency allows the pharmacist or physician to rapidly comprehend the nature of the error and take immediate corrective action, drastically improving workflow safety.

4.4.1 Case Example 1: Dopamine vs. Dobutamine

A physician verbally orders ”Dopamine”, but the system transcribes or a nurse types ”Dobutamine”. Both are cardiovascular agents but have vastly different indications (vasopressor vs. inotrope). The system computes the Jaro-Winkler distance and phonetic similarities, outputting a LASA probability of 0.98. The system instantly generates a severe alert, blocking the dispensing action.

4.4.2 Case Example 2: Metformin (Handling Safe Independent Contexts)

The system evaluates the string ”Metformin” against ”Mitomycin”. While orthographically somewhat similar, the Patient Context Validation engine checks the patient’s records. The patient has a verified diagnosis of Type 2 Diabetes (E11.9) and no oncological history. The system overrides the base string similarity and flags the input as clinically safe, preventing unnecessary alert fatigue.

4.5 Advantages over Traditional Detection Methods

- **Uniquely Robust Multi-Dimensional Checking:** By synthesizing orthographic, phonetic, and n-gram similarity metrics, the system vastly outperforms simple dictionary-based lookup tools. It dynamically adapts to newly synthesized drug names.
- **Actionable Alert Fatigue Suppression:** The integration of biological patient context reduces false positive alerts significantly, ensuring that when an alarm triggers, clinicians inherently trust its validity.
- **Audio-to-Text Pipeline Integration:** Incorporating cutting-edge speech transcription prevents errors at the very inception of the prescribing process, catching mispronunciations before they ever enter the digital text record.

CHAPTER 5

Final Findings And Conclusion

5.1 Summary of Key Epidemiological and Algorithmic Findings

The comprehensive execution of this advanced clinical decision support system successfully demonstrates the immense viability of artificial intelligence in mitigating fatal healthcare errors. By utilizing a 9-dimensional feature engineering pipeline that deeply analyzes string morphology and phonetic encoding, combined with the Whisper Speech-to-Text architecture, the system achieves an unprecedented 95.8% accuracy in detecting Look-Alike Sound-Alike (LASA) medication errors. Furthermore, the integration of clinical context validation effectively solves the persistent issue of alert fatigue that plagues legacy EHR systems.

5.2 Definitive Conclusion of Research Impact

This capstone endeavor explicitly confirms that static, rule-based medical safety systems are computationally insufficient for the modern hospital environment. The dynamic, probabilistic nature of ensemble machine learning classifiers provides a far more robust safety net. By deploying this system, clinical administration pathways can be safeguarded seamlessly, protecting patients from devastating biological harm caused by simple transcription or pronunciation errors.

5.3 Real-World Impact and Deployment Challenges

The real-world impact of deploying such a system is profound, potentially saving thousands of lives annually by intercepting catastrophic drug mismatches at the bedside. Furthermore, it significantly reduces the financial liability of healthcare institutions by preventing costly litigation associated with malpractice.

However, deploying this algorithmic pipeline into active clinical environments presents several challenges. Hospital IT infrastructures are notoriously rigid and bound by strict HIPAA compliance regulations. Integrating a Python-based machine learning API requires establishing secure, encrypted data tunnels to ensure patient diagnostic codes are parsed without violating privacy laws. Moreover, gaining clinician trust requires extensive usability training to ensure the system is viewed as a helpful aide rather than an intrusive administrative hurdle.

5.4 Scalability

To achieve national or global impact, the system must demonstrate high scalability. The current REST API architecture, built on FastAPI, is inherently stateless, allowing it to be easily dockerized and horizontally scaled. By deploying the inference engine across a load-balanced cloud infrastructure, the system can theoretically handle thousands of concurrent prescription validations per second, serving massive hospital networks without introducing unacceptable latency into the clinical workflow.

5.5 Current Technical Limitations

While highly accurate, the system currently relies on clear acoustic input for the Whisper model. Environments with extreme ambient noise (such as active trauma bays) may degrade the initial speech transcription accuracy. Additionally, the Patient Context Validation engine relies on the assumption that the patient's electronic health record is up-to-date with current ICD-10 diagnostic codes. If a diagnosis is missing, the system may conservatively revert to generating an alert, slightly increasing the false-positive rate.

5.6 Identified Future Scopes and Expansion Architectures

Future architectural enhancements will focus on integrating large language models (LLMs) to perform more nuanced semantic evaluations of free-text clinical notes. Furthermore, deploying the system using containerized Kubernetes clusters would allow for infinite scalability across multiple regional hospital networks, and incorporating Fast Healthcare Interoperability Resources (FHIR) standards will allow native, out-of-the-box integration with Epic and Cerner enterprise systems. Expanding the dataset to include international drug nomenclatures and multiple languages will further globalize the system's efficacy.

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